SHBT 261
GitHub repository: https://github.com/tzuhsi711/SHBT261_MiniProject1_CeceJen
Name: Tzu-Hsi (Cece) Jen
Deadline: March 3rd, 2026 11:59pm

Mini-Project 1: Image Classification with Caltech-101 Dataset

I. Introduction

Image classification has been greatly advanced by deep learning and machine learning models, which can automatically extract complex visual patterns and improve both accuracy and efficiency. [1] By reducing reliance on manual feature engineering, these approaches allow more scalable and robust analysis of large image datasets. However, differences in model architecture and training choices can strongly influence classification performance and generalisation. To examine these effects, this study compares a classical feature-based model, histogram of oriented gradients with support vector machine (HOG+SVM), with two convolutional neural network architectures, ResNet50 and EfficientNetB0, using the Caltech-101 dataset, which contains 102 image classes, and further evaluates how input image resolution and optimisation algorithm influence model performance.

HOG is a computer vision feature descriptor commonly used in object detection and image classification. It works by computing image gradients and grouping them into orientation bins. The image is divided into small cells, where histograms of edge directions are calculated, then normalised across neighbouring blocks to reduce the effects of illumination and contrast differences. By focusing on gradients, which capture changes in pixel brightness values across neighbouring regions, HOG extracts important shape and edge information while ignoring less useful image details. The normalised histograms from all blocks are then combined into a single feature vector representing the whole image, which is used by a classifier such as SVM to perform image classification. [2] However, manual feature extraction can be limited, as it is inherently constrained by predefined, handcrafted descriptors. [3] It is also time-consuming and heavily dependent on proficient human expertise. [4] CNNs overcome these limitations by automatically learning relevant features directly from images, enabling more powerful and scalable image classification. Pre-trained CNN models such as ResNet50 and EfficientNetB0 further enhance this approach by leveraging representations learned from large-scale image datasets. [3]

ResNet50 is a 50-layer convolutional neural network widely used for image classification due to its strong performance and ability to learn complex visual features. Its key innovation is the use of residual (skip) connections, which allow the input to bypass one or more layers and be added directly to the output. This design addresses the vanishing gradient problem in very deep networks by enabling residual learning, where the model learns the difference between the input and the desired output rather than the full transformation, improving gradient flow and training stability. [5] The architecture consists of stacked residual blocks followed by global average pooling, which summarises each feature map by averaging spatial dimensions while preserving channel depth, thereby reducing the number of parameters and helping mitigate overfitting [6]. A final softmax layer outputs class probabilities, and the class with the highest probability is selected as the prediction.

EfficientNetB0 is another CNN model that achieves high classification accuracy while remaining computationally efficient, and is widely used for image classification tasks. Its strong performance is driven by compound scaling, which systematically and proportionally adjusts the network's depth (number of layers), width (number of channels), and input resolution to optimise performance without unnecessary model complexity. The architecture is built using lightweight convolutional layers and inverted residual blocks, which expand feature channels and then compress them to capture meaningful information efficiently while keeping the number of parameters low. This design enables EfficientNetB0 to achieve strong classification performance with fewer parameters and lower

computational cost than many traditional CNNs, making it well suited for tasks that require a balance between accuracy and efficiency. [5]

In addition to model comparison, an ablation study was conducted to evaluate the classification performance of the best model under different input image resolutions (64×64 and 128×128) and optimisation algorithm, adaptive moment estimation (Adam) and stochastic gradient descent (SGD). Higher input resolution is expected to improve overall performance preserving finer and critical visual features that may be lost at lower resolution. [7,8] The choice of optimiser may also influence convergence behaviour and generalisation performance. Adam typically enables faster and more stable training through adaptive learning rates, while SGD may require more careful tuning but remains widely used for its simplicity and strong generalisation in many settings. [9,10]

Overall, this study evaluates the image classification performance of ResNet50, EfficientNetB0, and HOG+SVM using the Caltech-101 dataset. By evaluating model architecture alongside input resolution and optimisation strategy, it provides insights into how design and training decisions influence classification accuracy and generalisation.

II. Methods (description of models, dataset, ablations and metrics used)

Data preparation

The Caltech-101 dataset contains 102 image classes and was split into training (70%, 6,352 images), validation (15%, 1,326 images), and test (15%, 1,466 images) sets. Data augmentation was applied to the training set using random horizontal flipping, rotation (±0.05), and zoom (±0.10) to improve generalisation. Images were resized to 128×128 pixels for all models, with 64×64 additionally evaluated in an ablation study. CNN inputs were standardised using model-specific preprocessing functions, while images for the HOG+SVM model were rescaled to [0,1]. Given the large dataset with many classes, class labels were encoded as integer categories rather than one-hot vectors to provide a more compact representation for multi-class classification. Integer encoding stores a single class index per sample, whereas one-hot encoding represents each label as a 102-dimensional binary vector (one element per class), substantially increasing dimensionality. Using integer labels therefore reduces memory usage and computational overhead.

CNN Model: ResNet50

The first CNN was implemented using ResNet50 as the backbone with ImageNet pre-trained weights. The backbone was frozen so that only the newly added classification head was trained. Images were loaded using the Keras image dataset loader with integer-encoded class labels, providing a compact representation for multi-class classification and maintaining consistency with the HOG+SVM model for fair performance comparison. To transform convolutional feature maps into a compact representation, Global Average Pooling (GAP) 2D was applied to the backbone output. A Dropout layer (rate=0.3) was used to reduce overfitting by randomly deactivating 30% of neuron activations during training. The model was trained using the Adam optimiser (learning rate=3e-4). Final class probabilities were generated using a dense output layer with softmax activation.

CNN Model: EfficientNetB0

The second CNN used EfficientNetB0 as the backbone with ImageNet pre-trained weights, with the backbone similarly frozen so that only the classification head was trained. All other components were kept identical to the ResNet50 model to ensure a controlled comparison, allowing performance differences to be attributed solely to the feature extraction capabilities of the backbone architecture.

ML Model: HOG+SVM

HOG was used for feature extraction and classification was performed using a linear support vector classifier (LinearSVC), a linear implementation of the SVM. Images were converted to NumPy arrays, converted to grayscale, rescaled to [0,1], and resized to a fixed resolution before feature extraction. HOG features were computed using 128×128 images, 9 orientation bins, 8×8 pixels per cell, 2×2 cells per block, and L2-Hys block normalisation. These settings follow commonly adopted configurations in object recognition tasks and provide a balance between capturing local edge structure and computational efficiency [11].

To ensure that all feature dimensions contributed equally during training, the extracted HOG feature vectors were standardised using z-score normalisation before fitting the SVM classifier. The linear SVM was trained with a maximum of 50,000 optimisation iterations, which encourages model convergence, and a regularisation parameter C=1.0, which provides a balanced trade-off between maximising the margin and minimising training errors. Smaller values of C allow more tolerance to misclassification and produce a wider margin, which can lead to underfitting. Larger values of C enforce stricter separation of training samples with a narrower margin, which may increase the risk of overfitting. [12]

Ablations study

To evaluate the impact of input image resolution and optimiser choice on classification accuracy and generalisation, the best-performing model was further analysed in an ablation study. Model performance was compared across two input image resolutions (64×64, 128×128) and two optimisation algorithms (Adam and SGD).

Model evaluation

All models were evaluated using overall accuracy (training and test), top-5 accuracy, per-class accuracy, precision, recall, F1-score, and confusion matrices highlighting the 20 lowest-performing classes. Overall accuracy reflects general performance, while per-class accuracy and confusion matrices provide class-level insight. Top-5 accuracy was reported due to the large number of visually similar classes. Both macro and weighted precision, recall, and F1-score were calculated, with macro metrics treating classes equally and weighted metrics accounting for class imbalance.

III. Results (figures/tables of metrics + plots + ablations)

i. Overall accuracy

EfficientNetB0 achieved the highest test accuracy (0.9120), top-5 accuracy (0.9843), macro and weighted precision (0.9002, 0.9157), macro and weighted recall (0.8939, 0.9120), and macro and weighted F1-score (0.8896, 0.9088), outperforming both ResNet50 and the HOG+SVM model (Table 1). Across all models, training accuracy was higher than test accuracy (ResNet50: 0.9912 vs 0.8909; EfficientNetB0: 0.9817 vs 0.9120; HOG+SVM: 1.0000 vs 0.4386), indicating varying degrees of overfitting. Top-5 accuracy was consistently higher than top-1 test accuracy across models,

suggesting that many misclassifications involved visually similar classes, with the correct label frequently ranked among the top predictions.

Table 1. Performance comparison across ResNet50, EfficientNetB0, and HOG+SVM models. All models were trained with resolution of 128×128. ResNet50 and EfficientNetB0 were trained with Adam optimiser.

	Training accuracy	Test accuracy	Top-5 accuracy	Macro precision	Weighted precision	Macro recall	Weighted recall	Macro F1	Weighted F1
ResNet50	0.9912	0.8909	0.9775	0.8883	0.9014	0.8457	0.8909	0.8557	0.8893
EfficientNetB0	0.9817	0.9120	0.9843	0.9002	0.9157	0.8939	0.9120	0.8896	0.9088
HOG+SVM	1.0000	0.4386	0.5887	0.3585	0.4294	0.2334	0.4386	0.2517	0.3989

ii. Per-class accuracy

The top five classes with the lowest and highest per-class accuracy were identified for each model (Tables 2–3). For the lowest-performing classes, crocodile and lobster were consistently poorly classified by ResNet50 and EfficientNetB0. The HOG+SVM model exhibited markedly weaker class-level performance, with the five lowest-ranked classes all achieving zero classification accuracy. For the highest-performing classes, minaret consistently ranked among the top classes in EfficientNetB0 and the HOG+SVM models. While the highest-performing classes in both ResNet50 and EfficientNetB0 achieved perfect classification accuracy (1.000), the HOG+SVM model showed lower maximum per-class accuracy, with its best-performing classes remaining below 1.000.

Table 2. Top 5 classes with the lowest accuracy across ResNet50, EfficientNetB0, and HOG+SVM models.

ResNet50		EfficientNetB0		HOG+SVM	
Class	**Accuracy**	**Class**	**Accuracy**	**Class**	**Accuracy**
octopus	0.167	water_lilly	0.143	crab	0.000
lobster	0.286	lobster	0.286	dalmatian	0.000
schooner	0.500	BACKGROUND_Google	0.563	flamingo	0.000
chair	0.500	cannon	0.571	joshua_tree	0.000
crocodile	0.500	crocodile	0.625	cup	0.000

Table 3. Top 5 classes with the highest accuracy across ResNet50, EfficientNetB0, and HOG+SVM models.

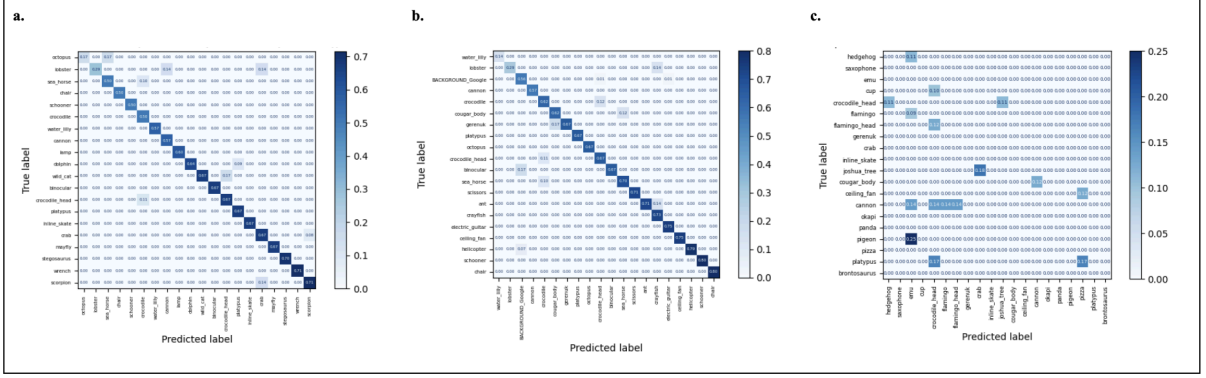
ResNet50		EfficientNetB0		HOG+SVM	
Class	**Accuracy**	**Class**	**Accuracy**	**Class**	**Accuracy**
saxophone	1.000	metronome	1.000	Motorbikes	0.992
laptop	1.000	minaret	1.000	Faces_easy	0.970
gerenuk	1.000	nautilus	1.000	airplanes	0.950
cellphone	1.000	flamingo	1.000	trilobite	0.929
camera	1.000	yin_yang	1.000	minaret	0.917

v. Confusion matrix and per-class accuracy

The confusion matrix displays the top 20 classes with the lowest per-class accuracy (Fig. 1a–c). Diagonal values represent correct classifications, while off-diagonal values indicate misclassifications. The EfficientNetB0 model exhibited stronger diagonal values and fewer off-diagonal confusions compared to the other models, consistent with its higher test accuracy and F1-scores, and reflecting more accurate and consistent predictions across classes. Across all models, several low-performing classes overlapped. In contrast, the HOG+SVM models exhibited sparse diagonal elements and scattered off-diagonal activations, indicating frequent misclassification and poor class separability.

SHBT 261
GitHub repository: https://github.com/tzuhsi711/SHBT261_MiniProject1_CeceJen
Name: Tzu-Hsi (Cece) Jen
Deadline: March 3rd, 2026 11:59pm

Fig. 1a-c Row-normalised confusion matrices showing the 20 lowest-performing classes across (a) ResNet50, (b) EfficientNetB0, and (c) HOG+SVM models.



vi. Ablation study

EfficientNetB0 was selected for the ablation study due to its best overall performance. Model behaviour was evaluated under two experimental conditions: input image resolution (64×64, 128×128) and optimiser type (Adam, SGD).

vi.i Input image resolution comparison

EfficientNetB0 achieved the best overall performance at a higher input resolution of 128×128, with higher training accuracy (0.9817), test accuracy (0.9120), top-5 accuracy (0.9843), macro and weighted precision (0.9002, 0.9157), macro and weighted recall (0.8939, 0.9120), and macro and weighted F1-score (0.8896, 0.9088) compared to the 64×64 model (Table 4). Across both resolutions, training accuracy remained higher than test accuracy, indicating some degree of overfitting. However, the performance gap between training and test accuracy was smaller at 128×128 than at 64×64, suggesting improved generalisation with higher-resolution inputs.

Table 4. Performance comparison of EfficientNetB0 across different input image resolutions.

	Training accuracy	Test accuracy	Top-5 accuracy	Macro precision	Weighted precision	Macro recall	Weighted recall	Macro F1	Weighted F1
64×64	0.8742	0.7490	0.9100	0.7109	0.7530	0.6703	0.7490	0.6704	0.7384
128×128	0.9817	0.9120	0.9843	0.9002	0.9157	0.8939	0.9120	0.8896	0.9088

The top five classes with the lowest and highest per-class accuracy were identified for each model (Table 5–6). For the lowest per-class accuracy, water_lilly and lobster were consistently poorly classified by both models. For the highest per-class accuracy, all top 5 highest-performing classes yielded perfect classification accuracy (1.000), with nautilus and yin_yang consistently ranked among the top categories in both models.

Table 5. Top 5 classes with the lowest accuracy across image resolutions for EfficientNetB0.

64×64		128×128	
Class	**Accuracy**	**Class**	**Accuracy**
ant	0.000	water_lilly	0.143
water_lilly	0.143	lobster	0.286
lobster	0.143	BACKGROUND_Google	0.563
wild_cat	0.167	cannon	0.571
platypus	0.167	crocodile	0.625

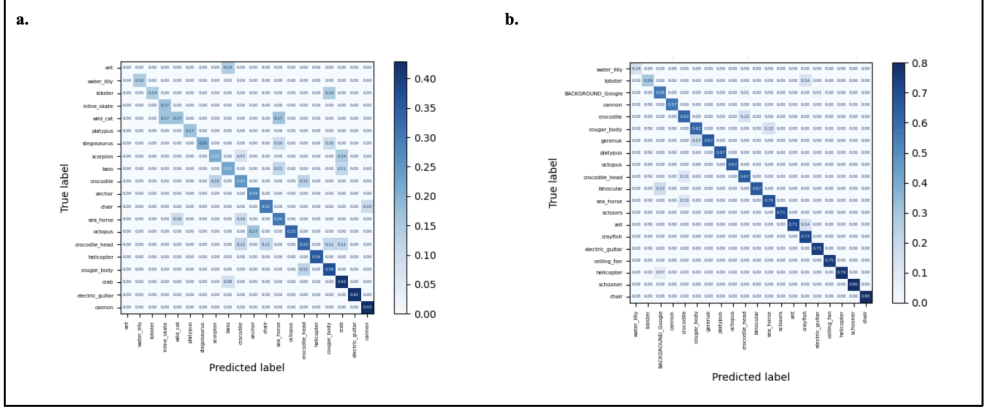
SHBT 261
GitHub repository: https://github.com/tzuhsi711/SHBT261_MiniProject1_CeceJen
Name: Tzu-Hsi (Cece) Jen
Deadline: March 3rd, 2026 11:59pm

Table 6. Top 5 classes with the highest accuracy across image resolutions for EfficientNetB0.

64×64		128×128	
Class	**Accuracy**	**Class**	**Accuracy**
soccer_ball	1.000	metronome	1.000
nautilus	1.000	minaret	1.000
garfield	1.000	nautilus	1.000
strawberry	1.000	flamingo	1.000
yin_yang	1.000	yin_yang	1.000

The confusion matrix displays the top 20 classes with the lowest per-class accuracy (Fig. 2a–b). At 128×128 resolution, the model exhibited substantially fewer off-diagonal confusions compared to 64×64 resolution, indicating improved classification performance.

Fig. 2a-b Row-normalised confusion matrices showing the 20 lowest-performing classes for the EfficientNetB0 model at (a) 64×64 resolution and (b) 128×128 resolution.



vi.ii Optimiser comparison

As EfficientNetB0 achieved the best overall performance at an input resolution of 128×128, this model was selected for optimiser comparison between Adam and SGD. EfficientNetB0 trained with the Adam optimiser outperformed SGD across all evaluation metrics (Table 7). Training accuracy remained higher than test accuracy for both models, indicating some degree of overfitting. Overall, Adam produced substantially better classification performance than SGD.

Table 7. Performance comparison of EfficientNetB0 across different optimisers.

	Training accuracy	Test accuracy	Top-5 accuracy	Macro precision	Weighted precision	Macro recall	Weighted recall	Macro F1	Weighted F1
Adam	0.9817	0.9120	0.9843	0.9002	0.9157	0.8939	0.9120	0.8896	0.9088
SGD	0.8097	0.7844	0.9359	0.7689	0.7791	0.6725	0.7844	0.6762	0.7550

The top five classes with the lowest and highest per-class accuracy were identified for each model (Table 8–9). For the lowest per-class accuracy, the Adam optimiser showed reduced performance for water_lilly (0.143) and lobster (0.286). In contrast, the SGD optimiser exhibited substantially lower per-class performance with zero classification accuracy. For the highest per-class accuracy, all top 5 highest-performing classes yielded perfect classification accuracy (1.000) under both optimisers. Classes including metronome, minaret, and yin_yang were consistently ranked among the highest-performing classes under both optimisers. Overall, Adam demonstrated stronger per-class performance compared to SGD.

SHBT 261
GitHub repository: https://github.com/tzuhsi711/SHBT261_MiniProject1_CeceJen
Name: Tzu-Hsi (Cece) Jen
Deadline: March 3rd, 2026 11:59pm

Table 8. Top 5 classes with the lowest accuracy across optimisers for EfficientNetB0.

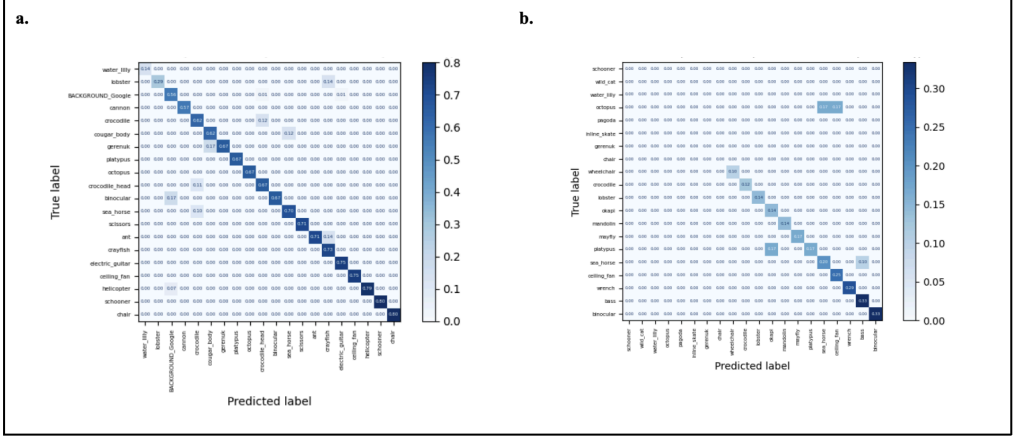
Adam		SGD	
Class	**Accuracy**	**Class**	**Accuracy**
water_lilly	0.143	schooner	0.000
lobster	0.286	octopus	0.000
BACKGROUND_Google	0.563	inline_skate	0.000
cannon	0.571	wild_cat	0.000
crocodile	0.625	pagoda	0.000

Table 9. Top 5 classes with the highest accuracy across optimisers for EfficientNetB0.

Adam		SGD	
Class	**Accuracy**	**Class**	**Accuracy**
metronome	1.000	metronome	1.000
minaret	1.000	minaret	1.000
nautilus	1.000	menorah	1.000
flamingo	1.000	rhino	1.000
yin_yang	1.000	yin_yang	1.000

The confusion matrix shows the 20 lowest-performing classes across optimisers for EfficientNetB0 (Fig. 3a–b). The Adam optimiser achieved correct classification for more classes, reflected by stronger diagonal values. In contrast, the SGD model showed limited correct classification across many categories, indicating weaker classification ability.

Fig. 3a–b. Row-normalised confusion matrices showing the 20 lowest-performing classes for the EfficientNetB0 model with (a) Adam optimiser and (b) SGD optimiser.



Discussion

This study compared deep learning and classical machine learning approaches for multi-class image classification on the Caltech-101 dataset. Overall, the CNN models outperformed the classical HOG+SVM approach, with EfficientNetB0 achieving the best overall performance. Performance varied with input resolution and optimiser, with the highest accuracy obtained using 128×128 image resolution and the Adam optimiser.

EfficientNetB0 demonstrated strong classification capability under these settings, achieving the highest test accuracy. Although training accuracy exceeded test accuracy, suggesting some degree of overfitting, the generalisation gap was smaller than in lower-resolution and SGD-trained models. High

precision indicates relatively few false positive predictions, while high recall reflects fewer false negatives. The elevated F1-score further demonstrates a balanced trade-off between precision and recall across classes.

In contrast, the HOG+SVM model exhibited substantially poorer classification performance, with test accuracy below 0.50, indicating markedly weaker performance than the CNN models. The sparse diagonal structure and widespread off-diagonal confusions observed in the row-normalised confusion matrix for the 20 lowest-performing classes illustrate its limited discriminative capacity and confusion among visually similar categories. This finding is consistent with known limitations of handcrafted feature descriptors, which rely on manually engineered gradient features and often require careful parameter tuning, making them less effective than deep feature representations learned by CNNs for complex, high-variability image datasets.

Across all models, top-5 accuracy consistently exceeded top-1 test accuracy. This suggests that many misclassifications involved visually similar categories, where the correct class was frequently ranked among the top predicted candidates. For precision, recall, and F1-score, weighted metrics were consistently higher than macro metrics, indicating stronger performance on dominant classes relative to less represented ones. Classes such as lobster and water_lilly repeatedly appeared among the lowest-performing categories. This pattern likely reflects class imbalance within the dataset. Prior to splitting, lobster contained 29 images and water_lilly 26 images, whereas the largest classes contained between 300 and 560 images. The small number of training samples in minority classes likely limited the model's ability to learn stable and generalisable feature representations.

Conclusion

Further optimisation strategies should be explored to reduce overfitting, improve generalisability, address class imbalance, and enhance overall classification performance. In addition to evaluating different input image resolutions and optimisers, hyperparameters such as batch size and learning rate could be tuned to improve training stability and convergence. [13,14,15] Regularisation techniques, including adjusting dropout rates and applying L1 and L2 penalties, may help control model complexity and reduce overfitting. [14] Incorporating batch normalisation could further stabilise the learning process and improve learning efficiency. [15] To address class imbalance, data augmentation with varied transformations or targeted sampling strategies could improve representation of minority classes and enhance classification performance across categories. [16,17]

Overall, this study demonstrates that both model architecture and optimisation strategy play a critical role in classification performance, reinforcing the importance of deep feature learning for robust, scalable, and accurate high-dimensional multi-class image recognition.

SHBT 261
GitHub repository: https://github.com/tzuhsi711/SHBT261_MiniProject1_CeceJen
Name: Tzu-Hsi (Cece) Jen
Deadline: March 3rd, 2026 11:59pm

Reference

1. Tran, Anh T et al. "Strategies to Improve the Robustness and Generalizability of Deep Learning Segmentation and Classification in Neuroimaging." *BioMedInformatics* vol. 5,2 (2025): 20. doi:10.3390/biomedinformatics5020020

2. Jain, Abhishek. "All about HOG (Histogram of Oriented Gradients)." *Medium*, 17 Dec. 2024, https://medium.com/@abhishekjainindore24/all-about-hog-histogram-of-oriented-gradients-869b5fab7bd5. Accessed 1 Mar. 2026.

3. Mbonu, Chinedu et al. "An Explorative Analysis of SVM Classifier and ResNet50 Architecture on African Food Classification." Department of Computer Science, Nazarbayev University and Nnamdi Azikiwe University.

4. Shaheed, Kashif et al. "EfficientRMT-Net-An Efficient ResNet-50 and Vision Transformers Approach for Classifying Potato Plant Leaf Diseases." *Sensors (Basel, Switzerland)* vol. 23,23 9516. 30 Nov. 2023, doi:10.3390/s23239516

5. Kansal, Kajal et al. "ResNet-50 vs. EfficientNet-B0: Multi-Centric Classification of Various Lung Abnormalities Using Deep Learning." *Procedia Computer Science* vol. 235 (2024): 70–80. doi:10.1016/j.procs.2024.04.007.

6. Yamashita, Rikiya et al. "Convolutional neural networks: an overview and application in radiology." *Insights into imaging* vol. 9,4 (2018): 611-629. doi:10.1007/s13244-018-0639-9

7. Kannojia, Suresh Prasad et al. "Effects of Varying Resolution on Performance of CNN Based Image Classification: An Experimental Study." *International Journal of Computer Sciences and Engineering* vol. 6,9 (2018): 451–456. doi:10.26438/ijcse/v6i9.451456.

8. Lopes, Ines et al. "On the impact of input resolution on CNN-based gastrointestinal endoscopic image classification." *Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference* vol. 2025 (2025): 1-4. doi:10.1109/EMBC58623.2025.11254367

9. Sayın, K. A. et al. "On the Synergy of Optimizers and Activation Functions: A CNN Benchmarking Study." *Mathematics* vol. 13,13 (2025): 2088. doi:10.3390/math13132088

10. Prabhat, Gyan, and Ashoke Kumar Mahato. "Adam vs. SGD: Achieving Comparable Generalization in Image Classification Through Adaptive Techniques." *IJCSPUB – International Journal of Current Science* vol. 15, no. 2 (April 2025): 273–279. Available at: https://rjpn.org/IJCSPUB/papers/IJCSP25B1033.pdf

11. N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05)*, San Diego, CA, USA, 2005, pp. 886-893 vol. 1, doi: 10.1109/CVPR.2005.177.

12. EITCA Academy. "What Is the Purpose of the C Parameter in SVM? How Does a Smaller Value of C Affect the Margin and Misclassifications?" 7 Aug. 2023. https://eitca.org/artificial-intelligence/eitc-ai-mlp-machine-learning-with-python/support-vector-machine/svm-parameters/examination-review-svm-parameters/what-is-the-purpose-of-the-c-parameter-in-svm-how-does-a-smaller-value-of-c-affect-the-margin-and-misclassifications/

13. Hargurjeet. *"7 Best Techniques to Improve the Accuracy of CNN W/O Overfitting: Dataset — CIFAR-10."* Medium, 27 May 2021, https://medium.com/

14. Darwish, D. (2024) 'Improving techniques for convolutional neural networks performance', *European Journal of Electrical Engineering and Computer Science*, 8(1). doi:10.24018/ejece.2024.8.1.596.

15. Albattah, Waleed, and Rehan Ullah Khan. "Impact of imbalanced features on large datasets." *Frontiers in big data* vol. 8 1455442. 13 Mar. 2025, doi:10.3389/fdata.2025.1455442

16. Wojciuk, Mikolaj et al. "Improving classification accuracy of fine-tuned CNN models: Impact of hyperparameter optimization." *Heliyon* vol. 10,5 e26586. 23 Feb. 2024, doi:10.1016/j.heliyon.2024.e26586

17. Vakalopoulou M, Christodoulidis S, Burgos N, et al. Deep Learning: Basics and Convolutional Neural Networks (CNNs) 2023 Jul 23. In: Colliot O, editor. Machine Learning for Brain Disorders [Internet]. New York, NY: Humana; 2023. Chapter 3. Available from: https://www.ncbi.nlm.nih.gov/books/NBK597497/ doi: 10.1007/978-1-0716-3195-9_3